

## Multiple Choice (10 marks)

1. A starvation-free job-scheduling policy guarantees that no job waits indefinitely for service. Which of the following job-scheduling policies is starvation-free?
  - (a) Youngest job first
  - (b) Least frequently used
  - (c) Round-robin
  - (d) Most recently used
  - (e) Longest job first
2. Let  $l = [1,2,3,4]$ . What is  $\min(l)$  in Python3?
  - (a) '1'
  - (b) 0
  - (c) -1
  - (d) 31,2,3,4
3. Which is the largest asymptotically?
  - (a)  $O(n^2)$
  - (b)  $O(n^3)$
  - (c)  $O(\sqrt{n})$
  - (d)  $O(2^n)$
  - (e)  $O(\log n)$
4. If the peak voltage value of a signal is 20 times the peak voltage value of the noise, what is the SNR? What is the  $\text{SNR}_{\text{dB}}$  (in 3 sig.fig.)?
  - (a) 15.0
  - (b) 13.0
  - (c) 30.0
  - (d) 34.0
  - (e) 26.0
5. Which of the following statements about a remote procedure call is true?
  - (a) It is used to call procedures with addresses that are farther than  $2^{16}$  bytes away.
  - (b) It cannot return a value.
  - (c) It cannot pass parameters by reference.
  - (d) It cannot call procedures implemented in a different language.

## True / False (10 marks)

*Answer True or False*

6. True or False: A convex 3D object with 15 vertices and 39 edges has 20 faces, which is a common misconception in polyhedral geometry.
7. True or False: In Python 3, the expression `b[::2]` returns the elements at odd indices, resulting in `[13, 17, 21]` from the list `b`.
8. True or False: The boolean expression `a[i] == max || !(max != a[i])` simplifies to `a[i] <= max`, indicating less than or equal to max.
9. True or False: A simple planar graph with 30 vertices can have a maximum of 84 edges according to Euler's formula.
10. True or False: Clustering techniques are effective in detecting fraudulent credit card transactions by identifying anomalous spending patterns.

## Long Form Response (20 marks)

*Answer each question with a clear and complete explanation.*

11. Write a function to check if the triangle is valid or not.
12. Write a function to sort the given array by using counting sort.

*End of Paper*